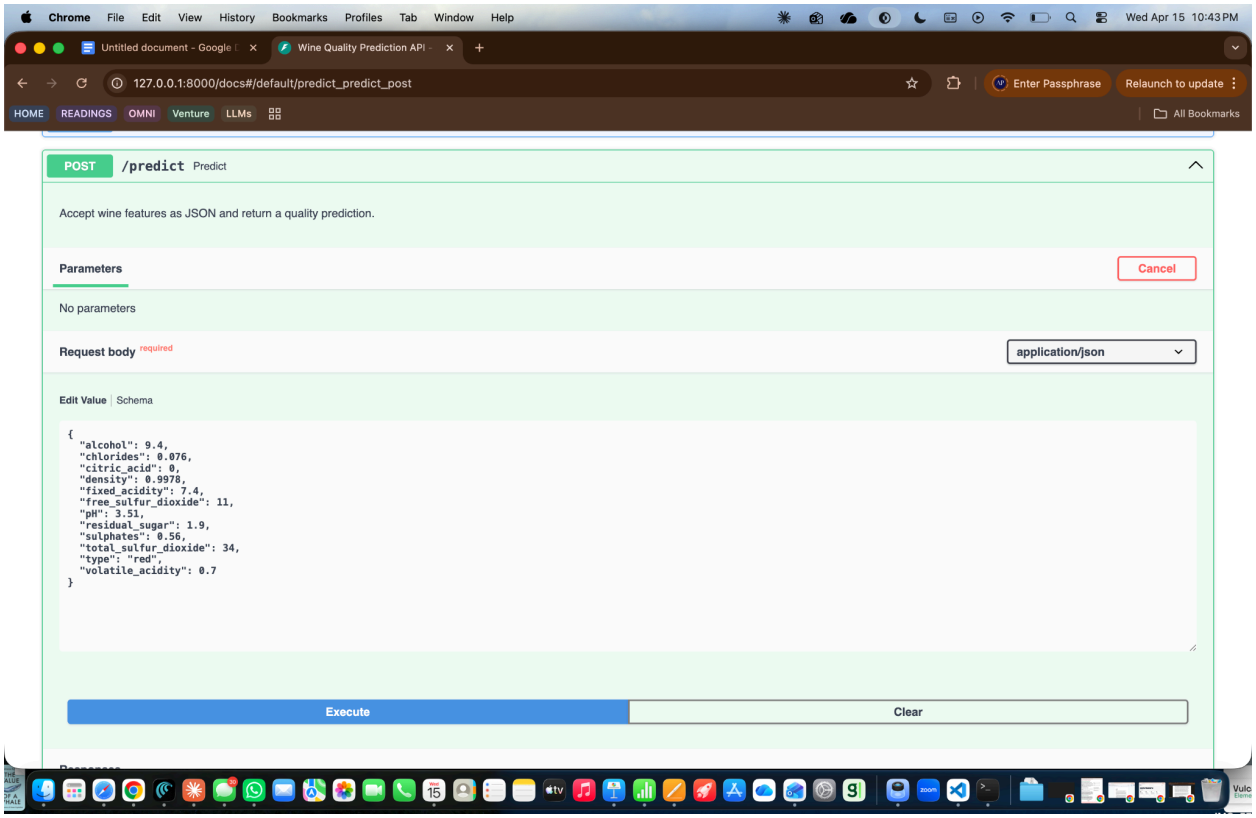


```
lightgbm  0.6332  0.4498  0.4530     1.791
gbc       0.5654  0.3440  0.3509     0.328
dt        0.5764  0.3709  0.3713     0.008
lda       0.5188  0.2739  0.2803     0.005
ridge     0.4688  0.2317  0.2504     0.005
lr        0.4869  0.2382  0.2526     0.197
qda       0.4842  0.2458  0.2481     0.005
knn       0.4577  0.1789  0.1807     0.008
dummy     0.2653  0.0000  0.0000     0.005
nb        0.4219  0.1511  0.1528     0.005
svm       0.3206  0.1031  0.1353     0.015
ada       0.3399  0.0877  0.0969     0.018

PyCaret Model Comparison (Top Results):
                                   Model  Accuracy     AUC  Recall   Prec.      F1   Kappa     MCC  TT (Sec)
et                   Extra Trees Classifier    0.6611  0.3431  0.6611  0.6665  0.6463  0.4683  0.4745     0.043
rf                 Random Forest Classifier    0.6602  0.3413  0.6602  0.6657  0.6450  0.4674  0.4733     0.058
xgboost            Extreme Gradient Boosting    0.6521  0.3315  0.6521  0.6478  0.6398  0.4611  0.4645     0.034
lightgbm  Light Gradient Boosting Machine    0.6453  0.3291  0.6453  0.6448  0.6332  0.4498  0.4530     1.791
gbc             Gradient Boosting Classifier    0.5843  0.0000  0.5843  0.5753  0.5654  0.3440  0.3509     0.328
dt                  Decision Tree Classifier    0.5771  0.2772  0.5771  0.5772  0.5764  0.3709  0.3713     0.008
lda             Linear Discriminant Analysis    0.5426  0.0000  0.5426  0.5207  0.5188  0.2739  0.2803     0.005
ridge                       Ridge Classifier    0.5366  0.0000  0.5366  0.4614  0.4688  0.2317  0.2504     0.005
lr                     Logistic Regression    0.5351  0.0000  0.5351  0.4913  0.4869  0.2382  0.2526     0.197
qda          Quadratic Discriminant Analysis    0.4808  0.0000  0.4808  0.4997  0.4842  0.2458  0.2481     0.005
knn                  K Neighbors Classifier    0.4722  0.2571  0.4722  0.4584  0.4577  0.1789  0.1807     0.008
dummy                     Dummy Classifier    0.4366  0.2000  0.4366  0.1906  0.2653  0.0000  0.0000     0.005
nb                           Naive Bayes    0.4022  0.2589  0.4022  0.4522  0.4219  0.1511  0.1528     0.005
svm                    SVM - Linear Kernel    0.3996  0.0000  0.3996  0.4143  0.3206  0.1031  0.1353     0.015
ada                   Ada Boost Classifier    0.3543  0.0000  0.3543  0.4219  0.3399  0.0877  0.0969     0.018

Best model: ExtraTreesClassifier
Transformation Pipeline and Model Successfully Saved
Model saved as best_pipeline.pkl

============================================================
SCIKIT-LEARN WORKFLOW (Manual: ExtraTreesClassifier)
============================================================

Classification Report (Scikit-Learn):
              precision    recall  f1-score   support

           3       0.00      0.00      0.00         6
           4       0.50      0.14      0.22        43
           5       0.76      0.72      0.74       428
           6       0.66      0.80      0.73       567
           7       0.68      0.57      0.62       216
           8       0.93      0.36      0.52        39
           9       0.00      0.00      0.00         1

    accuracy                           0.70      1300
   macro avg       0.50      0.37      0.40      1300
weighted avg       0.70      0.70      0.68      1300


✅ discovery.py complete!
Artifacts generated:
  - pycaret_comparison.csv
  - Confusion Matrix.png
  - best_pipeline.pkl
  - sklearn_classification_report.txt
(wine) alexiospoyiadjis@Mac wine-quality-ml %
```